

Internship Write Up - VISA
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About Visa :

Visa Inc. (*/ˈviːzə, ˈviːsə/*; stylized as *VISA*) is an American multinational financial services corporation headquartered in San Francisco, California. It facilitates electronic funds transfers throughout the world, most commonly through Visa-branded credit cards, debit cards and prepaid cards. Visa is one of the world's most valuable companies.

Visa was founded in 1958 by Bank of America (BoFA) as the BankAmericard credit card program. In response to competitor Master Charge (now Mastercard), BoFA began to license the BankAmericard program to other financial institutions in 1966. By 1970, BoFA gave up direct control of the BankAmericard program, forming a consortium with the other various BankAmericard issuer banks to take over its management. It was then renamed Visa in 1976.

VISA has achieved the zenith by becoming the world's leader in digital payments technology. The company strives to accomplish its mission to connect the world through the most creative, reliable and secure payment network - enabling individuals, businesses, and economies to thrive. Visa provides secure and reliable payments around the world, and is capable of handling more than 65,000 transaction messages a second with Visanet. Millions of users are now able to travel as far as they want, even overseas with a single card that can be used with ease and comfort for all transactions. Visa has instilled this feeling of trust and reliability in one and all. The company perseverently and tirelessly works towards the dream of transforming the lives of people through technology. It primarily focuses on building a cashless future for everyone, everywhere.

Job Profile Description As Given By The Company :

Job Profile : Software Engineering Intern

Location : Bengaluru, Karnataka, India

What is Visa?

WE ARE NOT A BANK, WE ARE NOT A CREDIT CARD COMPANY!!

Visa is the world's leader in digital payments technology!

At Visa, your individuality fits right in. Working here gives you an opportunity to impact the world, invest in your career growth, and be part of an inclusive and diverse workplace. We are a global team of disruptors, trailblazers, innovators and risk-takers who are helping drive economic growth in even the most remote parts of the world, creatively moving the industry forward, and doing meaningful work that brings financial literacy and digital commerce to millions of unbanked and underserved consumers.

You're an Individual. We're the team for you. Together, let's transform the way the world pays. Behind the Visa brand are our talented employees who continuously raise the bar with innovative solutions and products that deliver the convenience and security of digital currency to more people all over the world. We are currently looking for candidates for our 2021 Summer Internship Program. Visa interns gain real-world experience working for one of the world's most recognized brands.

During this intensive 10-12 week internship, you will work on challenging and rewarding assignments, gain exposure to Visa business leaders and executives, and participate in social and networking opportunities.

So what are we looking for in a Software Engineering Intern?

- HTML, CSS, Javascript, Responsive Design, Accessibility, Objective C, and/or Android SDK
- Training in C/C++ and Object Oriented design is preferred
- Familiarity with software development methodology that is used to structure, plan, and control the process of software development
- Experience with user research and user centered design.
- Experience with usability (e.g., as a tester, analyst, or developer using user-centered data)
- Experience with accessibility (Section 508 and/or WCAG 2.0) a plus
- Knowledge of user interface design as applied to Web-based applications and mobile devices. • Knowledge of both Macs and PCs helpful

- Proficiency in MS Excel, Word and PowerPoint is required

The below skills will help us differentiate you with your peer group :

- Strong ability to collaborate
- Highly motivated, resourceful and ability to work independently with initial guidance and direction
- Teamwork, interpersonal & relationship-building skills, and ability to lead by influence and example
- Project management skills and attention to detail
- Strong written and oral communication, including large-group presentations
- Proven ability to prioritize and manage multiple projects simultaneously

Candidate Profile Framing :

The key four areas to be touched upon for building a resume are :

- **Technical Knowledge/Skillset/Interests :**
 - Talk about the technical knowledge and skills you possess i.e the various programming and scripting languages, tools, etc that you know and the various database management systems, operating systems, etc that you are familiar with.
 - You can also mention the domains you are passionate about like AI-ML, Computer Vision, Data Science, etc.
- **Projects :**
 - Projects play a salient role in defining your candidature in any professional environment. This section can drastically strengthen or weaken your profile. Thus, it is important to do innovative and unique projects and include them in your resume.
 - Include all the projects that you have done so far till then and the ongoing projects too.
 - Ensure your project implements a significant coding concept, be it data structures, web development or domain-centric projects on AI,ML or Data Science.
 - Choose product-based project ideas that are usable and can be deployed.
 - Build a project that solves a real time problem.
 - If you did the project as a team, do mention it.
- **Internships (if any) :** If you have undertaken any internships in a corporate firm in prior, do mention about that. Include details about your role or job profile, etc.
- **Courses :** You can mention here about the online certified courses that you have undertaken through platforms like Coursera, Udemy, NPTEL, etc.
- **Non-Technical Skillset/Interests, Soft Skills :**
 - Here, you can enunciate upon your skills outside of your IT/CS domain.
 - You can mention other activities you are interested in like sports, writing, art, etc.
 - You can also stress upon your soft skills like professional writing, creative writing, speaking/orations, etc.

Preparation Strategy For Competitive Programming:

- Starting off your preparation for Competitive Programming (CP) early is a big advantage. You get to explore and learn slowly and an early starter gets to cover a vast expanse of problems.
- For CP, first you have to find your niche i.e find your perfect programming language that you are strong at by analyzing your strengths and liabilities while solving problems using various languages. Explore thoroughly before you make a choice as shifting programming languages amidst your preparation might create havoc.
- It is advised to stick to the order of recommended languages below :
 - C++/Python (Highly Recommended, especially C++)
 - Java (Also a good choice)
 - C
- My choice of language was Python as I really admire and revere the language and I was accustomed to it since high school. Learning a new language like C++ from scratch was tough for me. Python was comfortable and I never faced any issues with that. But a majority of people recommend C++. Java is a good choice as well. On the other hand, C is quite less preferred.

- CP will take you to a completely different realm of coding and programming. CP requires a problem solving approach which would be entirely different from the normal monotonous coding problems with more remembering involved that you solve pertaining to the syllabus courses.
- The biggest change begins from a small step. So start off by solving a few simple problems with difficulty level 'easy' from the sites given below :
 - Hackerrank
 - Leetcode
 - Geeksforgeeks
- Develop a conceptual learning approach towards programming rather than a theoretical approach. Focus more on building ideas rather than just getting the output by typing the code.
- Before jumping into the perspective of coding it out, channelise your focus to crack the logic. After the logic is figured out, coding it out is easy. Getting a strong foothold on CP is possible only when you have a vehement flow of ideas as a culmination of ideas and logic forms the basic foundation of CP.
- Gradually bring in the time component as given below and see if you are able to solve a problem with the target :
 - Easy : 5 Minutes
 - Medium : 15 Minutes
 - Hard : 30 Minutes
- If you are unable to, strive to achieve it by strengthening your practice.
- Slowly advance to higher difficulty levels and try solving more problems.
- Then, gradually start expanding various concepts of CP as per the following order :
 - Arrays
 - Strings
 - Searching and Sorting Algorithms
 - Stacks And Queues
 - Linked Lists
 - Trees
 - Graphs
 - DP
 - Back-tracking
 - Bit Manipulation
- The only key to unlock the skill of CP is rigorous practice. Consistently practice and solve more and more problems. Attempt as many contests as you can to gain exposure. Explore and solve as much as you can.
- You may not be able to always work on a daily basis given the tight schedules, but being consistent in solving problems is what matters. Keep getting back to solving the next day even if you miss a day. Long gaps will prove to be a great impediment to your flow of practice.
- And, it is not advised to use solutions readily available on the internet. However, after you solve, you can refer to a few sites (only trusted authentic sites) to see if your solution can be improvised.

Interview Preparation :

The above guidelines focus on how to prepare consistently on a daily basis. But the guidelines below throw light on how to prepare for the approaching upcoming interviews :

General Preparation :

- Revise the important concepts in CP. Practice the problems you feel are important and frequently asked and refresh the approaches used.
- Take a glance at important concepts of significant courses in our syllabus like Data Structures, OOPs, Algorithms, OS, DBMS to prepare yourself to answer technical questions.

Company Specific Preparation :

- Go through and practice the past years' questions of the particular companies from the archives.
- Websites For Archives :
 - Geeksforgeeks
 - Interviewbit

- Refer the archives for the coding, technical and aptitude questions asked in online assessments and interview rounds.
- Do a detailed research and study about the company.

Selection Process :

Round 1 : Online Coding Assessment :

Duration : 1 hour

No. of Qs : 2

Question 1 :

Concept Covered : Arrays

Gist of The Question :

It was a very simple and direct team formation question. Given an array of strengths of various members, form teams as pairs of people with the same strengths. The program finally returns the no. of teams that can be possibly formed.

Approach :

- Take the frequencies of each element and divide by two (floor division) and add them.
- To ensure no repetitions of iterations for same array elements, convert the python list into a set and compute frequencies.

Estimated Time : Less Than 5 Minutes

No. of Test Cases Passed : All 15/15

Difficulty Level : Easy (In My Opinion)

Question 2 :

Concept Covered : Simple Logical Computations With Good Geometry Visualization

Gist of The Question :

Given the cartesian coordinates of 3 vertices of a triangle ABC and 2 points P and Q on the cartesian plane.

We had to find which one of these 5 cases is true for the given input :

1. The vertices A, B and C cannot form a triangle.
2. The vertices A, B and C can form a triangle and the points P and Q both lie inside the triangle.
3. The vertices A, B and C can form a triangle and the points P and Q lie inside and outside of the triangle respectively.
4. The vertices A, B and C can form a triangle and the points P and Q lie outside and inside of the triangle respectively.
5. The vertices A, B and C can form a triangle and the points P and Q both lie outside the triangle.

The function should return 5 different integers corresponding to those 5 cases.

To Find If Vertices A, B and C Can Form A Triangle :

Approach :

- Compute the distance of the edges using the distance formula.
- Take the 3 possible combinations of pairs of edges.
- The sum of the lengths of those 2 edges should be greater than the third edge.

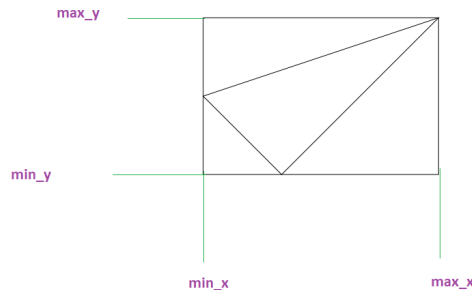
To Find If A Point P Lies Inside The Triangle ABC :

Approach 1:

- Find the minimum and maximum x and y coordinate among A, B and C.
- If x coordinates of P are lesser than the minimum x coordinate or greater than the maximum x coordinate or if similarly y coordinates of P are lesser than the minimum y coordinate or greater than the maximum y coordinate, then P is not situated inside the triangle.

Estimated Time For Approach 1 : Less Than 5 Minutes

Mistake in Approach 1 :



This concept of min and max draws a rectangle that inscribes the triangle which is wrong as P may also lie inside the rectangle and outside the triangle.

Approach 2:

- Compute the equations of the 3 lines AB, BC and AC using their coordinates in the form $ax+by=c$.
- Substitute P's x and y in the LHS and see if it is lesser or greater than c.
- Considering the points A, B and C are in clockwise direction, LHS should be greater than c for AC and lesser for AB and BC.

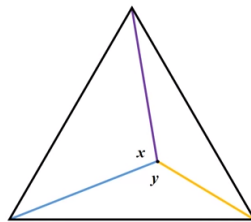
Estimated Time For Approach 2: 10 Minutes

No. of Test Cases Passed After Approach 2 : All 15/15

Difficulty Level of The Question : Medium (In My Opinion)

After the assessment, on resolving, A Better Approach :

- We can compute the area of a triangle with the 3 vertices' coordinates.
- Find the areas of ABC, ABP, ACP and BCP.
- If the sum of areas of ABP, ACP and BCP equals the area of ABC, then P lies inside tri.ABC.



Round 2 : Interview (Technical + HR) : [14 Student Shortlisted]

Platform : Microsoft Teams

[We were asked to turn on our videos]

Introductory Interaction Questions Asked:

- Tell me about yourself.
- Highlight your technical and non-technical skillset.
- Enunciate your strengths and weaknesses.

Tip :

It is advised to prepare for these questions and frame an answer in advance as you will face these questions in any interview you take up.

Resume - Based Questions :

- I was asked to explain any one completed project from my resume (I explained my DSA project).
- Questions : What did you learn from the project? & What problems did you face while working on it?

Coding Questions : (No. of Questions : 2)

How the Coding Round Was Carried :

- A hackerrank code pair link was given to us where we can type our code and obtain the output. The interviewer will have a live view of whatever we code.
- A question was given to us and we were asked to solve it quickly without explaining.

Tip :

Usually it is advised to be interactive while coding out your solution. Always think aloud and discuss your flow of ideas with the interviewer and they will help you out when you are stuck as well. Also, any mistake or mishap in your logic will be spotted in advance and you can change it then and there without wasting time. But, for us, due to the time constraint of only 1 hour for both technical and HR questionnaires.

- The code that we put forth should be optimized.
- Once we were done, we should report it to the interviewer.
- Then they would ask you to explain the code and answer the follow up questions.

Question 1 :

Concept Covered : Matrices, Binary Search

Tip : Always be thorough with commonly used searching and sorting algorithms.

Given an $m \times n$ matrix, we have to search for a value k .

Each row of the matrix is sorted and the last element of a row is always less than the first element of the next row.

Approach 1 :

- Populate all the entries to a 1D array of size mn .
- Perform binary search.

[I was told that converting to a 1D array is not allowed] So, to solve , Perform 1 binary search to find the row. Then, Perform another binary search to locate the element k in that row.
(Requires 2 Loops)

Follow Up Q : Can this be done without using 2 loops? Ans : No, since we are not allowed to convert into 1D, there should definitely be 2 searches, one to find the row and one more to locate the element in that row.

Question 2 :

Concept Covered : Arrays

The simple and very familiar two-sum question. To find pairs of integers in an array that yield a sum k .

Approach :

- Convert the array to python set.
- Traverse the set for each element i .
- If $k-i$ is also in i , add the key-value pair $(i, k-i)$ to a dictionary.
- Print the pairs.

[Tip : In this interview, we had no questions from technical concepts like DSA, DBMS, OS, etc. But they are sure to be asked in a majority of the interviews and cannot be neglected.]

Aptitude Question :

- Q : You have 2 ropes. Each rope will burn completely in 30 mins. Measure a time duration of 45 minutes using the 2 ropes.
- So, here we definitely burn one rope entirely to measure 30 minutes.
- The catch is to find how to measure 15 mins with the second rope.
- My Answer : It is 15 minutes when the rope burns half its length.
- [But there is no instrument to measure half
- My Answer : Folding the rope to get midpoint.
- [But the rope doesn't stay homogenous. It doesn't burn at a uniform rate throughout]
- After thinking for a minute, My Answer : Burn the rope at both ends (Irrespective of rate of burning, lighting it at 2 ends doubles the speed at which the matter in the rope burns, Speed doubled is time halved. So 15 mins can be measured) [Right Answer]

Company - Based Questions :

- Why VISA? (Tip : As mentioned earlier, one should be aware of the company to answer such questions.)
- Do you have any questions for us? (Tip : Here, do not say that you have no questions. Do ask something about the company that you are curious to know about. [Eg: I asked about the security mechanisms used in VISA that have formed a foundation to instill a sense of trust and reliability to millions of users across the globe.])

General Tips To Be Kept In Mind During The Interview.

- Be calm and poised throughout the interaction.
- Speak fluently without fumbling and put forth your points with confidence.

Final Result : Selected! [11 Students Selected From AU]

Feel free to reach me for doubts at..

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Overall Message :

This phase may be tough. You may face rejections despite acing the interview rounds. Never see this rejection as a factor that pulls you down. Stay motivated and never lose hope. A vehement learner gets gratified in the experiences and doesn't blindly expect good outcomes. Keeping that in mind, enjoy the course of learning and keep striving tirelessly towards your goal. Achievers never quit until they achieve the impossible. Remember always that the best awaits you. Keep up your perseverance in learning more and more and that indeed paves the way for you to triumph.

Hope this was useful to you! Wish you all the best for your preparation! You are sure to hit it up the park and ace your interviews!